Standby Generator

Model: FG-5087Z

OPERATION AND MAINTENANCE MANUAL

Version: 3.9

Date: November 2025

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This manual is organized into sections covering safety, technical specifications, installation, operation, maintenance, troubleshooting, and parts information. Each section should be reviewed before beginning work.

This equipment must be installed and operated in accordance with local codes, regulations, and industry standards. Consult with qualified personnel if you have questions about compliance requirements.

All personnel involved in installation, operation, or maintenance of this equipment must be properly trained and familiar with the contents of this manual.

The Generator is designed for industrial applications requiring reliable performance and long service life. Proper installation, operation, and maintenance are essential for achieving optimal results.

Before operating the FG-5087Z, ensure that all installation requirements have been met and that the equipment has been properly commissioned by qualified personnel.

This document contains detailed technical specifications, installation procedures, operational guidelines, and maintenance schedules to ensure optimal performance and safety.

For technical support or additional information, please contact your local distributor or visit our website.

Note

Always verify that the FG-5087Z is properly grounded according to local electrical codes before operation.

For more information see:

- *Installation Guide - Electrical Connections*

Warning	Description	Elimination/Action
⚠	Allow equipment to cool before maintenance. Use appropriate protective equipment when handling hot components.	Wait for equipment to reach ambient temperature. Wear heat-resistant gloves.
⚠	Some lubricants and materials may cause allergic reactions in sensitive individuals.	Use protective gloves. Ensure adequate ventilation. Consult material safety data sheets.
⚠	Pressure can build up during operation. Sudden release can cause injury.	Release pressure slowly using appropriate procedures. Never remove caps or plugs under pressure.
⚠	Overfilling can cause leakage, overheating, and equipment damage.	Follow manufacturer specifications for fill levels. Check levels regularly.
⚠	Mixing incompatible oils can cause equipment failure and void warranty.	Use only specified lubricants. Drain completely before changing oil type.
	Warming oil before use improves flow and ensures proper distribution.	Heat oil to recommended temperature (typically 40-50°C) before application.
	The amount of new lubricant should match the volume drained to maintain proper levels.	Measure drained volume accurately. Add equivalent amount of new lubricant.

Warning	Description	Elimination/Action
⚠	Residual oil can be slippery and cause falls. Dispose of properly.	Clean up spills immediately. Use appropriate absorbent materials. Follow local disposal regulations.

Environmental conditions such as temperature, altitude, and humidity can affect equipment performance. Specifications are based on standard conditions unless otherwise specified.

Some specifications may be optional or configurable. Consult with technical support or your local distributor to determine available options for your specific requirements.

All specifications are subject to change without notice as improvements are made to the product. Always refer to the latest version of this manual for current specifications.

This section provides detailed technical specifications for the Generator model FG-5087Z. All specifications are based on standard operating conditions unless otherwise noted.

General Specifications

Parameter	Value	Unit	Notes
Power Output	10	kW	As specified
Voltage Output	400V	V	Standard
Frequency	60	Hz	Optional
Fuel Consumption	40		
Weight	3567	kg	Approximate
Operating Temperature	-17 to 51	°C	Continuous
Storage Temperature	-20 to 65	°C	
Humidity	19 to 87	% RH	Non-condensing
Protection Class	IP66		IEC 60529
Noise Level	62	dB(A)	At 1m distance
Vibration	≤ 5	mm/s	RMS
Service Factor	1.15		

Dimensions

Dimension	Value	Unit	Tolerance
Length	1493	mm	±2 mm
Width	215	mm	±2 mm
Height	622	mm	±2 mm
Mounting Holes	M12		

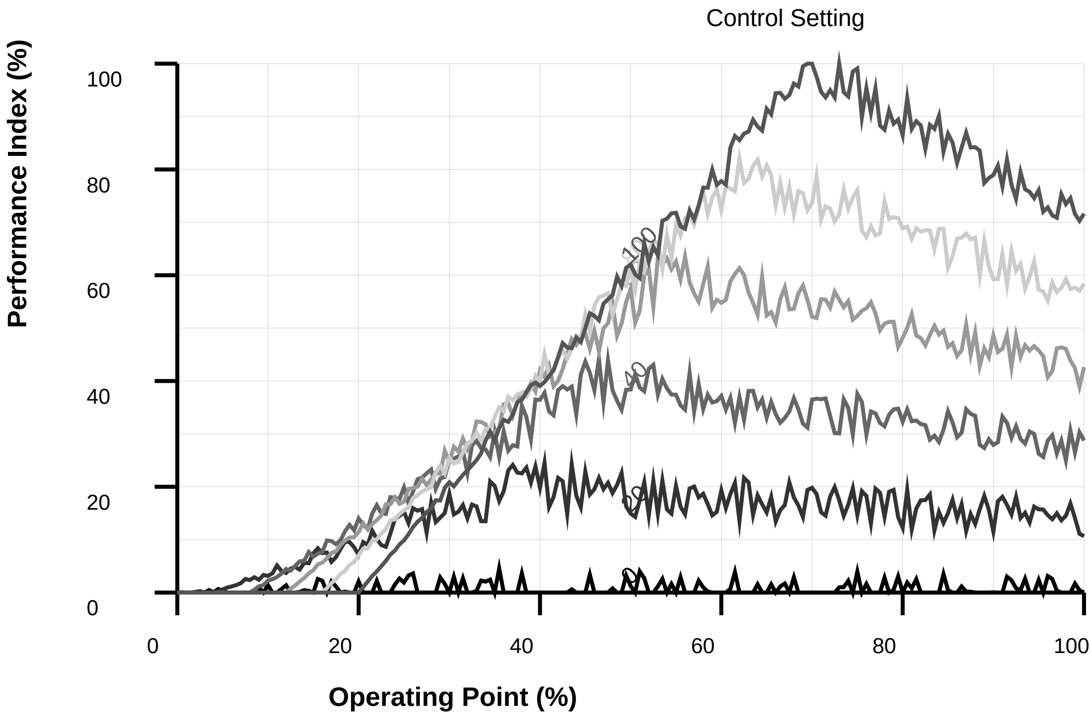
Dimension	Value	Unit	Tolerance
Shaft Diameter	34	mm	±0.1 mm
Base Height	184	mm	±1 mm

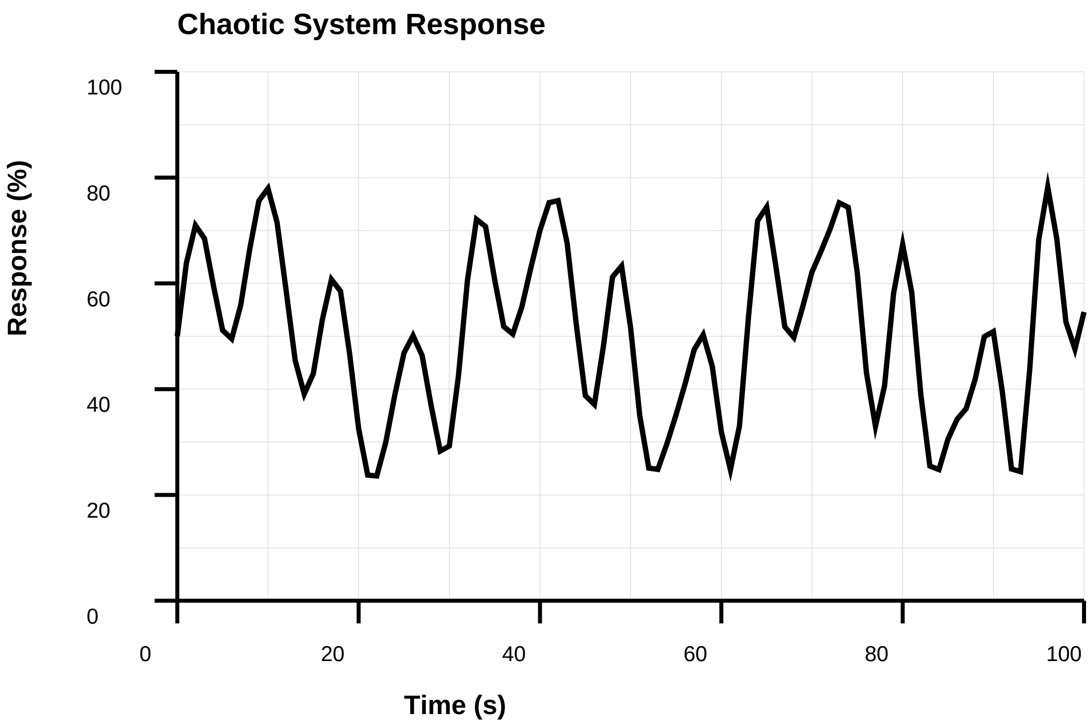
Performance Data

Operating Condition	Minimum	Nominal	Maximum	Unit
Power Consumption	90	100	105	%
Efficiency	85	92	98	%
Load Factor	23	76	105	%
Speed	742	1472	2954	rpm
Torque	11	71	169	Nm

Environmental Conditions

Condition	Operating Range	Storage Range	Unit
Temperature	-13 to 53	-22 to 65	°C
Relative Humidity	15 to 85	9 to 95	%
Altitude	0 to 1198	0 to 2698	m
Vibration	≤ 2	≤ 10	mm/s





Electrical Specifications

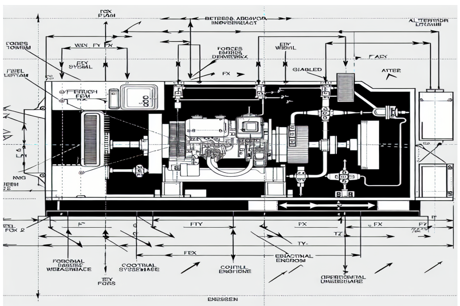
Parameter	Value	Unit	Standard
Supply Voltage	400V	V AC	IEC 60038
Frequency	60 Hz	Hz	IEC 60034
Current Rating	31	A	
Power Factor	0.92		
Starting Current	491	% of rated	
Insulation Class	F		IEC 60085
Protection	IP66		IEC 60529

Regular inspection of electrical connections and components is essential for safe and reliable operation. Include electrical inspection in your maintenance schedule.

When operating at non-standard conditions, consult with technical support to determine if derating or special considerations are required.

4.1 Installation Requirements

Installation Procedures

Action	Note
• Verify alignment before final assembly. • Check that all mounting surfaces are clean and free from burrs. • Use proper alignment tools as specified in the installation guide.	Misalignment can cause premature wear and failure. Refer to alignment procedures in Section 4.
• Apply proper torque to all fasteners. • Use torque values specified in the technical data sheet (39 Nm for main bolts). • Tighten in the correct sequence as shown in the installation diagram.	Overtightening can damage components. Undertightening can cause loosening during operation. 
• Lubricate the sealing with grease just before fitting. • (Not too early - there is a risk of dirt and foreign particles adhering to the sealing.) • Fill 2/3 of the space between the dust lip and the main lip with grease. • If the sealing is without dust lip, just lubricate the main lip with a thin layer of grease.	Ensure that no grease is applied to the marked surface. Use only approved lubricants specified in the parts list.

Action	Note
• Inspect the shaft surface before mounting. • If scratches or damage are found, the shaft must be replaced since it may result in future leakage. • Do not try to grind or polish the shaft surface to get rid of the defect.	Ensure proper surface finish is maintained. Refer to technical specifications for surface roughness requirements.

Auxiliary systems such as cooling, lubrication, and ventilation must be properly connected and verified before startup.

The Generator requires careful handling during transport and installation to prevent damage to critical components.

The installation site must meet all specified requirements for foundation, space, environmental conditions, and access for maintenance. Verify all requirements before proceeding.

Before beginning installation, review all safety instructions in Section 2 and ensure that all personnel involved are properly trained and qualified.

Some installation tasks may require specialized tools or equipment. Ensure all necessary tools are available before beginning.

Before installation, ensure the following requirements are met:

Installation Checklist

Requirement	Specification	Status
Adequate space for installation and maintenance access	As per installation manual	■
Proper foundation or mounting surface	As per installation manual	■
Correct electrical supply voltage and frequency	As per installation manual	■
Appropriate environmental conditions (temperature, humidity)	As per installation manual	■
Compliance with local building and electrical codes	As per installation manual	■

Foundation Requirements

Parameter	Minimum	Recommended	Unit
Foundation Thickness	327	321	mm
Concrete Strength	C22	C27	
Reinforcement	As per local code	As per local code	
Anchor Bolt Size	M20	M20	

Parameter	Minimum	Recommended	Unit
Anchor Bolt Depth	218	282	mm

4.2 Installation Steps

Installation Procedure

Step	Description	Duration	Tools Required
1	Unpack the equipment and inspect for damage.	15 min	Torque wrench, Wrench set
2	Position the equipment according to layout drawings.	30 min	Torque wrench, Level
3	Secure the equipment to the foundation using appropriate fasteners.	30 min	Torque wrench, Level
4	Connect electrical wiring according to wiring diagrams.	30 min	Torque wrench, Level
5	Connect auxiliary systems (cooling, lubrication, etc.).	15 min	Alignment tools, Torque wrench, Wrench set
6	Perform initial alignment and calibration.	2 hours	Torque wrench, Level, Alignment tools
7	Verify all connections and safety systems.	30 min	Alignment tools, Torque wrench, Wrench set
8	Perform initial test run according to startup procedure.	2 hours	Torque wrench, Alignment tools, Wrench set

Torque Specifications

Component	Bolt Size	Torque	Unit
Mounting Bolts	M16	97	Nm
Shaft Coupling	M8	41	Nm
Terminal Connections	M6	14	Nm
Cover Bolts	M8	16	Nm

Document all installation measurements, adjustments, and test results. This information is valuable for troubleshooting and future maintenance.

Verify that all required tools, equipment, and materials are available before beginning installation. Missing items can cause delays and safety hazards.

5.1 Operating Procedures

Before operating the equipment, ensure that all installation and commissioning procedures have been completed and that the equipment is in proper working condition.

During operation, be alert for unusual sounds, vibrations, or other signs of problems. Stop the equipment immediately if safety concerns arise.

Coordination with other equipment and systems may be required for optimal operation. Ensure proper communication and coordination procedures are in place.

Emergency procedures must be understood by all operators. Practice emergency shutdown procedures regularly to ensure quick response when needed.

Some operating modes may have specific requirements or limitations. Review operating mode specifications before changing modes.

Follow these procedures for safe and efficient operation:

Operating Parameters

Parameter	Setting	Range	Unit
Startup Sequence	Automatic	Manual/Automatic	
Operating Mode	Continuous	Continuous/Intermittent	
Control Method	Manual	Manual/Automatic/Remote	
Monitoring	Enabled	Enabled/Disabled	
Speed Setpoint	1442	1160	rpm
Load Limit	91	110	%

Startup Sequence

Step	Action	Duration	Status Indicator
1	Power On	Immediate	Power LED
2	System Initialization	11	Init LED
3	Self-Test	16	Test LED
4	Ready State	Continuous	Ready LED
5	Start Command	On Demand	Run LED

5.2 Control Functions

Control Functions

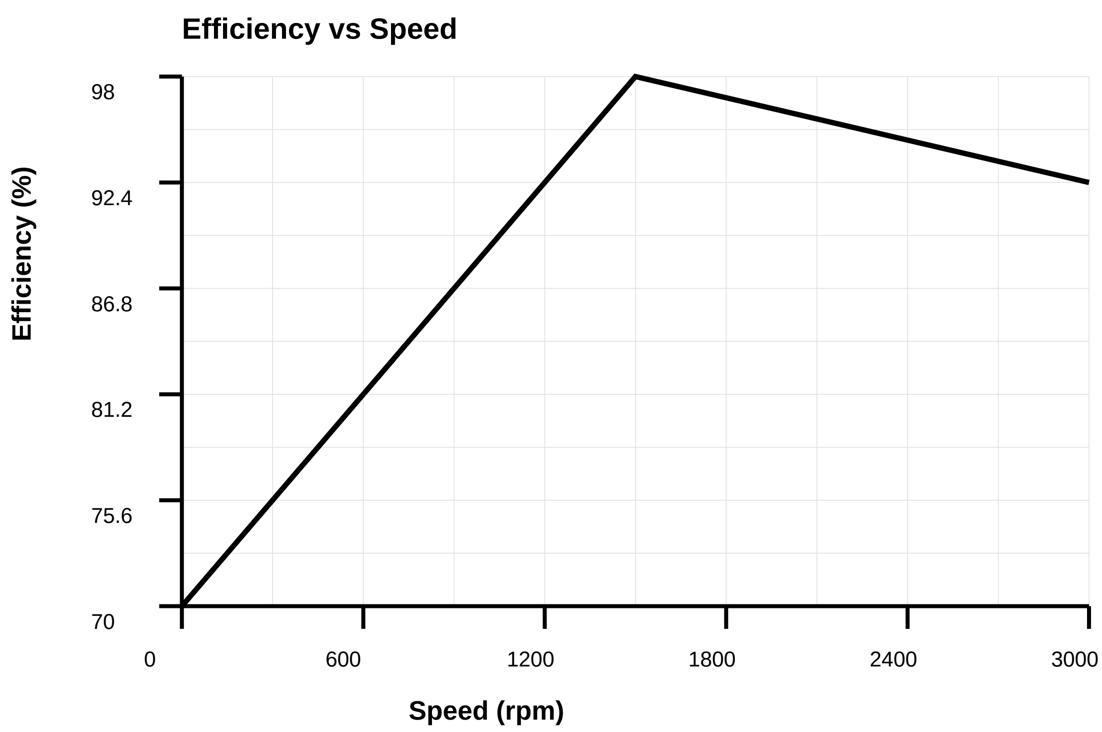
Control	Function	Location	Type
Start/Stop	Controls equipment operation	Panel A	Push Button

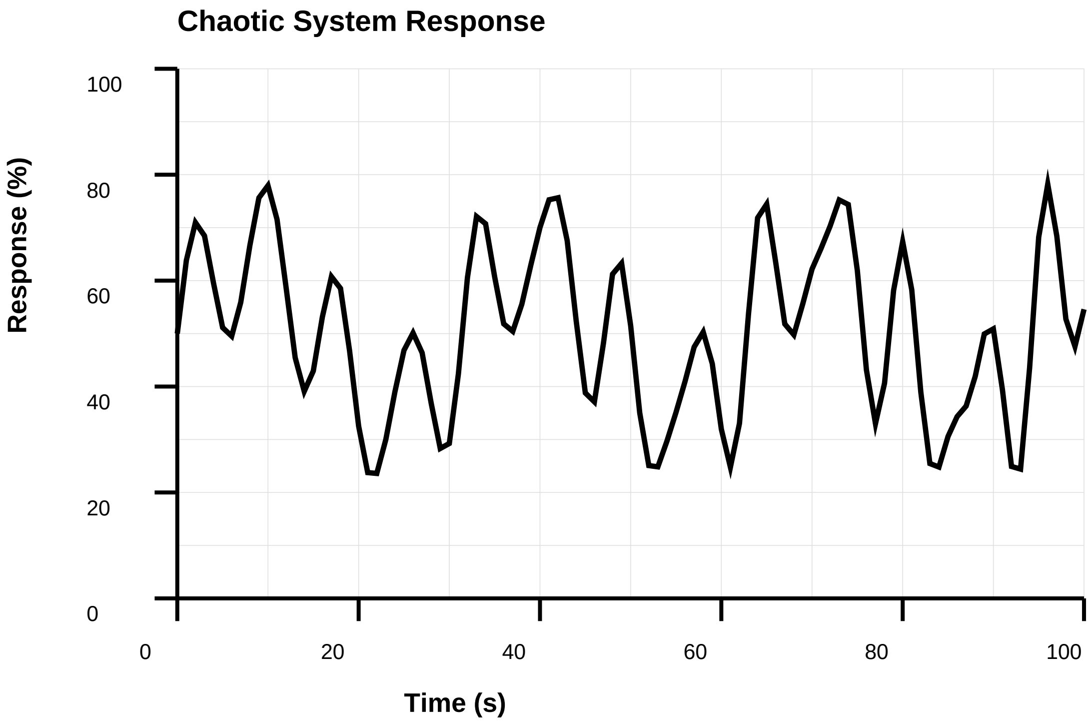
Control	Function	Location	Type
Speed Control	Adjusts operating speed within specified range	Panel B	Selector Switch
Emergency Stop	Immediately stops all operations	Panel C	Emergency Stop
Reset	Clears fault conditions and resets system	Panel D	Reset Button
Mode Selection	Selects operating mode	Panel E	Selector

Status Indicators

Indicator	Color	Meaning	Action
Power	Green	System powered	Normal operation
Run	Green	Equipment running	Normal operation
Fault	Red	Fault detected	Check fault code
Warning	Yellow	Warning condition	Review warning
Maintenance	Yellow	Maintenance due	Schedule service

Operating Characteristics





When operating in automatic mode, verify that all control signals are functioning correctly and that the equipment responds appropriately to commands.

Monitor operating parameters continuously during operation. Significant deviations from normal values may indicate problems requiring attention.

Emergency stop functions must be tested regularly to ensure they are functioning properly. Never disable safety systems.

6.1 Maintenance Schedule

Follow the maintenance schedule strictly to prevent premature failures and ensure optimal performance. Maintenance intervals may need to be adjusted based on operating conditions.

Before performing any maintenance, ensure the equipment is safely shut down and all safety protocols are followed. Refer to Section 2 for safety instructions.

All maintenance work must be performed by qualified personnel familiar with the equipment and maintenance procedures. Improper maintenance may cause equipment damage or safety hazards.

Regular maintenance is essential for maintaining the performance, reliability, and service life of the Generator. This section provides maintenance schedules and procedures.

Maintenance Schedule

Maintenance Task	Frequency	Duration	Personnel
Visual Inspection	Daily	5 min	Operator
Lubrication	Every 424 hours	30 min	Technician
Filter Replacement	Every 827 hours	1 hour	Technician
Bearing Inspection	Every 1222 hours	2 hours	Engineer
Complete Overhaul	Every 9497 hours	8 hours	Service Team
Calibration Check	Every 2590 hours	1 hour	Engineer

Inspection Checklist

Item	Check	Acceptance Criteria	Frequency
Bearings	Visual/Temperature	No noise, temp < 70°C	Weekly
Seals	Visual	No leakage	Weekly
Mounting	Visual/Torque	Bolts tight, no movement	Monthly
Electrical	Visual/Measurement	No damage, correct voltage	Monthly
Cooling	Visual/Temperature	Clean, adequate flow	Weekly
Vibration	Measurement	< 5 mm/s RMS	Monthly

6.2 Maintenance Procedures

- Always shut down and lock out equipment before maintenance.
- Follow manufacturer's recommendations for lubricants and spare parts.
- Inspect all components for wear, damage, or contamination.
- Replace worn or damaged parts immediately.
- Keep maintenance records for warranty and service history.
- Use only approved replacement parts and materials.

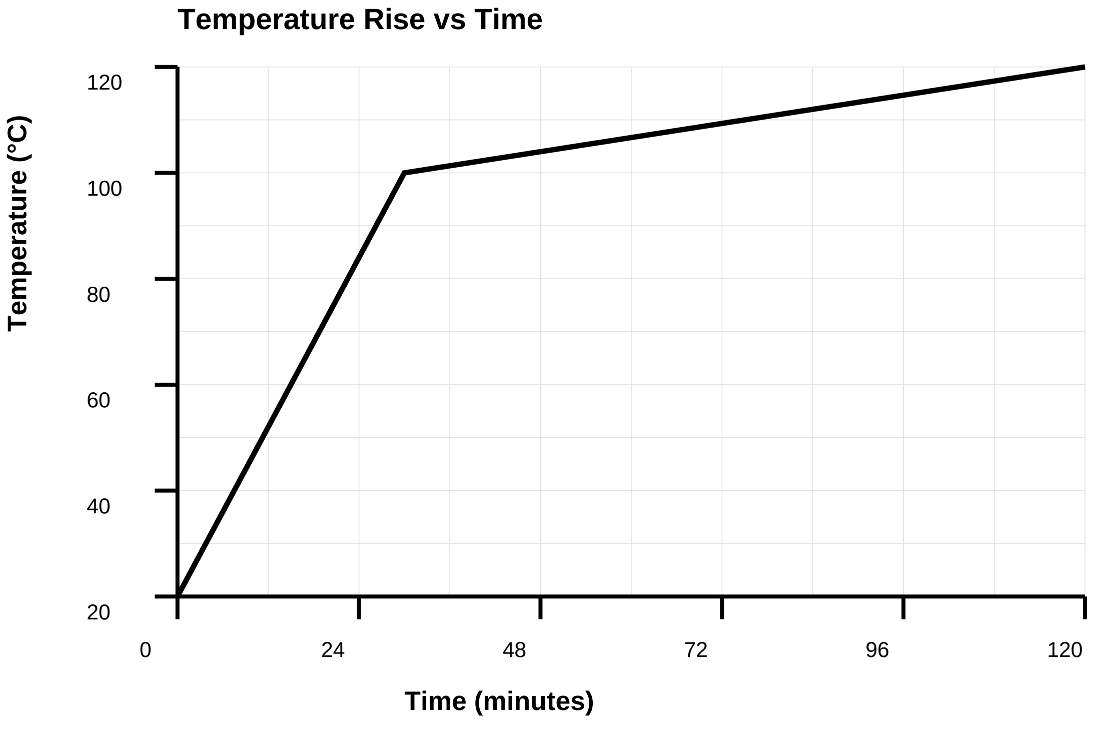
Component Maintenance Intervals

Component	Inspection Interval	Replacement Interval	Condition
Bearings	Every 1538 hours	Every 11515 hours	Based on condition
Seals	Every 731 hours	Every 3781 hours	If leaking
Filters	Every 610 hours	Every 1665 hours	If dirty
Belts	Every 1975 hours	Every 3407 hours	If worn

Some maintenance tasks require specialized tools or training. Do not attempt maintenance beyond your qualification level. Contact qualified service personnel when needed.

> **Note**
> Maintenance personnel must be properly trained and familiar with the equipment. Inexperienced personnel should work under supervision.
>
> For more information see:
> - *Safety Manual - Lockout/Tagout Procedures*

Temperature Characteristics



Before beginning troubleshooting procedures, ensure the equipment is safely shut down and all safety protocols are followed. Always refer to the safety instructions in Section 2 before performing any diagnostic work.

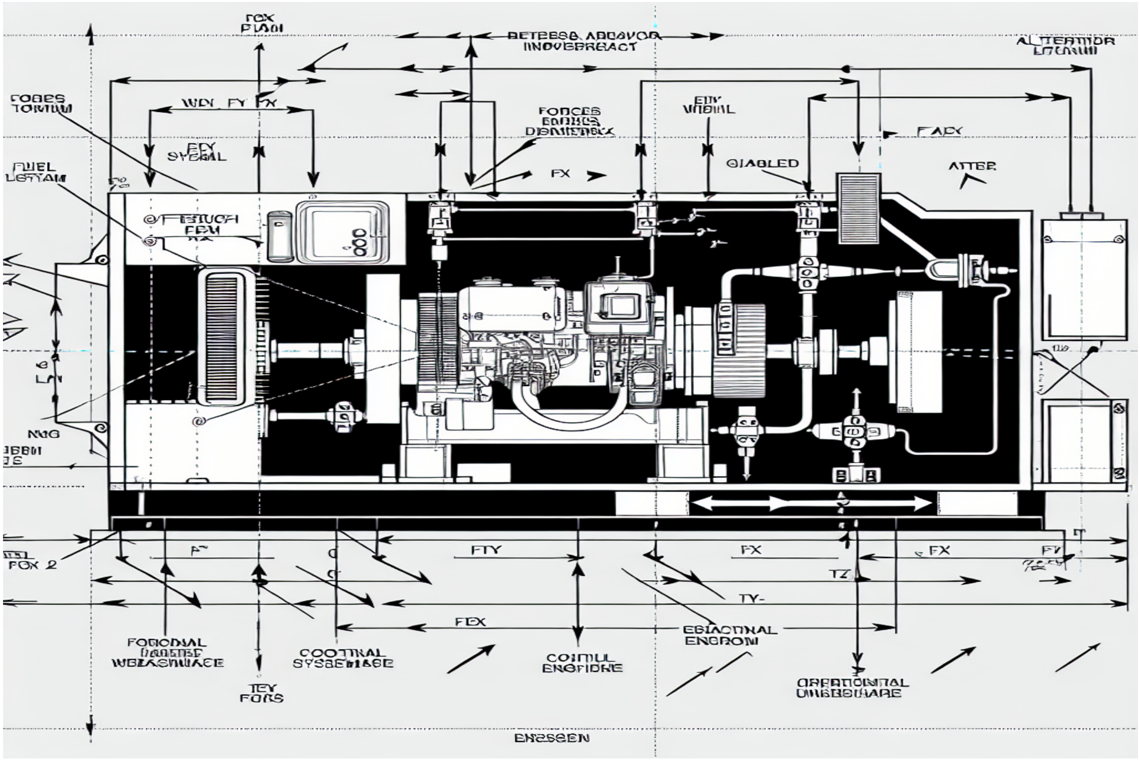
Many issues can be resolved through basic inspection and maintenance. However, some problems may indicate more serious conditions that require professional service. When in doubt, contact technical support.

Troubleshooting the Generator requires systematic diagnosis of symptoms and their underlying causes. This section provides guidance for identifying and resolving common operational issues.

Common Issues and Solutions

Symptom	Possible Cause	Solution
Fault Code G849: Engine fault	Engine failure, sensor issue, or control fault	Check engine, test sensors, verify control
Excessive noise at specific frequencies	Resonance, mechanical wear, or structural issue	Identify frequency, check for resonance, inspect mechanical components
Fault Code G755: Low voltage	Engine speed low, AVR fault, or load issue	Check engine speed, test AVR, verify load
Premature component failure	Incorrect operation, poor maintenance, or environmental factors	Review operating procedures, improve maintenance schedule, check environmental conditions
Motor overload trip	Excessive load, mechanical binding, or motor fault	Check load conditions, inspect for binding, test motor windings
Irregular speed or torque output	Control system calibration or feedback sensor issue	Recalibrate control system, verify feedback sensors, check signal integrity
Encoder feedback error	Encoder failure, cable issue, or mounting problem	Check encoder connections, verify signal integrity, inspect mounting
Bearing temperature high	Insufficient lubrication, bearing wear, or misalignment	Check lubrication levels, inspect bearings, verify alignment
Excessive vibration	Imbalance, misalignment, or worn bearings	Perform balancing check, verify alignment, inspect bearings
Leakage of fluid or lubricant	Damaged seals, gaskets, or overfilling	Inspect seals and gaskets, verify fill levels, replace damaged components
Unusual noise during operation	Bearing wear, misalignment, or loose components	Inspect bearings for wear, check alignment, tighten all fasteners

When troubleshooting, always start with the simplest possible causes before investigating more complex issues. Many problems can be resolved through basic inspection and cleaning.



Note

Always verify that the equipment is properly powered and all safety systems are functioning before attempting troubleshooting procedures.

For more information see:

- *Technical Support - Contact Information*

Spare Parts List

Part Number	Description	Quantity	Material	Supplier Part No.
FG-5087Z-183	Main Housing	4	Stainless Steel	SP-73442
FG-5087Z-302	Rotor Assembly	3	Cast Iron	SP-77954
FG-5087Z-476	Stator Assembly	3	Steel	SP-94676
FG-5087Z-600	Bearing Set	3	Cast Iron	SP-72940
FG-5087Z-205	Shaft	3	Aluminum	SP-23907
FG-5087Z-259	Seal Kit	1	Bronze	SP-98957
FG-5087Z-114	Gasket Set	1	Plastic	SP-64797
FG-5087Z-807	Mounting Bracket	3	Cast Iron	SP-14596
FG-5087Z-937	Cooling Fan	3	Stainless Steel	SP-66174
FG-5087Z-234	Terminal Box	1	Steel	SP-55661

Part Number	Description	Quantity	Material	Supplier Part No.
FG-5087Z-761	Nameplate	1	Aluminum	SP-27476
FG-5087Z-179	Fastener Kit	2	Plastic	SP-44520
FG-5087Z-630	Thermal Protection	4	Steel	SP-14542
FG-5087Z-672	Vibration Sensor	1	Plastic	SP-56576
FG-5087Z-295	Oil Filter	1	Stainless Steel	SP-74231
FG-5087Z-136	Cooling Coil	3	Steel	SP-75132
FG-5087Z-773	Impeller	3	Cast Iron	SP-40689

Recommended Spare Parts

Part Number	Description	Recommended Stock	Lead Time
FG-5087Z-492	Bearing Set	1-2	2-4 weeks
FG-5087Z-908	Seal Kit	1-2	2-4 weeks
FG-5087Z-130	Gasket Set	2-3	2-4 weeks
FG-5087Z-466	Filter Element	3-5	2-4 weeks
FG-5087Z-936	Thermal Fuse	3-5	4-6 weeks

Lubrication Requirements

Component	Lubricant Type	Quantity	Interval
Main Bearings	Oil ISO VG 68	75	Every 1883 hours
Gearbox	Oil ISO VG 220	3	Every 3522 hours
Seals	Silicone Grease	17	Every 832 hours

Refer to the wiring diagrams provided in the appendix. All connections must be made according to local electrical codes and regulations.

Terminal Connections

Terminal	Function	Wire Size	Voltage
L1, L2, L3	Power Input	4 mm²	400V
PE	Protective Earth	2.5 mm²	0V
U, V, W	Motor Output	6 mm²	Variable
24V	Control Supply	0.75 mm²	24V DC
GND	Control Ground	0.75 mm²	0V

The Generator is compliant with applicable international standards for industrial machinery and electrical equipment. In case of deviation from these standards, these are listed in the declaration of incorporation. The declaration of incorporation is part of the delivery.

Machine Standards

Standard	Description
ISO 12100	Safety of machinery - General principles for design - Risk assessment and risk reduction
IEC 60204-1	Safety of machinery - Electrical equipment of machines - Part 1: General requirements

Other Standards Used in Design

Standard	Description
IEC 61000-6-2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
IEC 61000-6-4	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
ISO 13849-1:2015	Safety of machinery - Safety related parts of control systems - Part 1: General principles for design

A. Technical Data Sheets

Additional technical data sheets and detailed specifications are available upon request.

Revision History

Revision History

Revision	Description
V	• Published in release R17.2. The following updates are made in this revision: • Clarified and added information in mounting instructions for rotating sealings. • Added information about optimization and calibration procedures in calibration chapter. • Replaced article number and name of grease, previously 3793289-2.
Z	• Published in release R21B. The following updates are made in this revision: • Clarified and added information in mounting instructions for rotating sealings. • Added information about optimization and calibration procedures in calibration chapter. • Replaced article number and name of grease, previously 3793289-2.

Revision	Description
N	• Published in release R24A. The following updates are made in this revision: • Updated information about special treated screws. • Removed information about position switches as they are no longer available. • Updated torque specifications for various components.
U	• Published in release R23A. The following updates are made in this revision: • Added sections in general procedures. • Safety restructured. • Updated spare part number brake release board (was DSQC555, is DSQC1010). • Information about business portal added.
Y	• Published in release R20B. The following updates are made in this revision: • New tool for replacement of parallel bar. • Updated maintenance procedures. • Added calibration requirements.
T	• Published in release R19B. The following updates are made in this revision: • Added sections in general procedures. • Safety restructured. • Updated spare part number brake release board (was DSQC555, is DSQC1010). • Information about business portal added.
W	• Published in release R24D. The following updates are made in this revision: • New tool for replacement of parallel bar. • Updated maintenance procedures. • Added calibration requirements.

B. Contact Information

Contact Information

Department	Contact
Technical Support	support@forgis.com
Sales	sales@forgis.com
Service	service@forgis.com
Emergency	+1-800-XXX-XXXX